DSAA3073: Theories in Data Science

Tutorial 9A: From Continuous Decisions to a Telescoping Regret Bound

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-17

Explorative projection

LESSON BACKBONE

How can we compete with a fixed-point benchmark in an infinite convex set without listing every candidate?

Tutorial 8B ended with a comparator theorem over a finite simplex and a familiar gradient-method template. An online decision may instead be any point in a continuous convex set. Listing a fine grid is tempting, but a grid neither preserves every feasible point nor proves that geometry and computation have become cheap.

We will keep the same choose-before-reveal comparator idea and replace enumeration by a geometric proof. Convexity first turns a loss gap into one gradient inner product. We will then define the feasibility correction that makes a raw gradient move usable, and squared distance will telescope across rounds.

The three core blocks take **40, 45, and 65 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge. The course's mirror-descent connection is not part of this Tutorial; it remains isolated to the answered zero-minute appendix after Tutorial 9B's complete core and course handoff.

Clock	Mathematical work	Protected time
0-40	Formulate online convex optimization and compute continuous-comparator regret	40 min core
40-55	Scheduled break	15 min
55-100	Turn convex loss gaps into feasible distance changes	45 min core
100-165	Execute projected OGD and telescope its regret proof	65 min core
165-225	Questions, reconstruction, alternative calculations	60 min buffer

Keep the comparator, replace the list

Tutorial 9A receives exactly four items from the verified Chapter 8 boundary. Reconstruct them before introducing any online convex-optimization notation.

No.	Available item	Permitted use at entry
1	external regret	Compare cumulative learner loss with one best fixed decision on the same realized loss sequence.
2	linear MWU regret theorem	Recognize a finite-simplex comparator result with a legal positive learning rate.
3	convex function	Use the chord inequality and its first-order consequence once differentiability is stated.
4	gradient-method template	Recognize an additive local move $x - \eta g$; no online update or feasibility theorem is inherited.

IN-CLASS CHECK

Prompt: Write the four entry items in order. Which item supplies a comparator, and which item supplies only a proposed local move?

Hint: Separate an online performance definition from an offline optimization template.

Answer:

> The order is external regret; the linear MWU regret theorem; convex function; gradient-method template. External regret supplies the best-fixed same-sequence comparator. The gradient-method template supplies only the additive proposal $x - \eta g$; it does not yet say when the point is feasible or what regret it achieves.

Now replace two experts by two assets. A portfolio is a point $x = (x_1, x_2)$ in the simplex

$$\mathcal{K} = \{x \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 = 1\}. \tag{1}$$

The three rows below have already been generated by legal commitments. The current loss vector is revealed only after the portfolio in that row has been chosen. Complete the protocol order and both cumulative-loss columns.

Round	chosen x_t	revealed loss g_t	$g_t^T x_t$	$g_t^T(u, 1-u)$
1	$\left(\frac{1}{2}, \frac{1}{2}\right)$	$(1, 0)$		
2	$\left(\frac{1}{4}, \frac{3}{4}\right)$	$(0, 1)$		
3	$\left(\frac{3}{4}, \frac{1}{4}\right)$	$\left(\frac{1}{2}, 0\right)$		
sum				

Card A	Card B	Card C
choose x_t from the feasible set	reveal the whole convex loss f_t	incur $f_t(x_t)$ and retain f_t for the comparator ledger
order:	order:	order:

Claim card	Accept?	Repair or reason
a fine grid is exactly the continuous simplex		
no explicit d means a dimension-free bound		
every continuous-set operation costs linear time		

Table 1: The continuous-comparator ledger. The student preserves the online information order before the general protocol is named.

IN-CLASS CHECK

Prompt: Complete the ledger. Which fixed portfolio minimizes the three-round loss, and what is the learner's regret? Then repair the three claim cards.

Hint: Write a fixed portfolio as $(u, 1-u)$ and minimize only after summing all three revealed losses.

Answer:

The learner losses are $\frac{1}{2}$, $\frac{3}{4}$, and $\frac{3}{8}$, totaling $\frac{13}{8}$. The fixed portfolio loses $u + (1-u) + \frac{u}{2} = 1 + \frac{u}{2}$, minimized at $u = 0$ with value 1. The regret is therefore $\frac{13}{8} - 1 = \frac{5}{8}$. A finite grid is only an approximation to the simplex; omitting an explicit d does not prevent geometric parameters from scaling with dimension; and the cost of operating on a continuous set depends on its representation rather than obeying one universal linear-time rule.

Definition: Online convex optimization and its registered objects

Let $\mathcal{K} \subseteq \mathbb{R}^d$ be a nonempty closed convex decision set and let the allowed losses be bounded convex functions $f_t : \mathcal{K} \to \mathbb{R}$. On round t, the learner chooses $x_t \in \mathcal{K}$ before f_t is revealed, then incurs $f_t(x_t)$. The revealed loss may be chosen adversarially and may depend on the committed action.

For the realized sequence $f_1, \dots, f_T$, regret against the fixed-point infimum benchmark is

$$R_T = \sum_{t=1}^{T} f_t(x_t) - \inf_{x \in \mathcal{K}} \sum_{t=1}^{T} f_t(x). \tag{2}$$

Thus $\mathcal{K}$ is the registered decision set and f_t is the registered round loss. The benchmark ranges over points that are fixed across rounds and uses the same revealed sequence as the learner. We write a minimum and speak of a **best fixed point** only when this infimum is attained.

The portfolio table is a concrete instance with linear losses $f_t(x) = g_t^T x$. In general, the loss need not be linear. The objects that do not change are the choose-before-reveal order, the infimum over fixed comparators in $\mathcal{K}$, and the same loss sequence on both sides of the regret expression. The dimension d, the geometry of $\mathcal{K}$, and the scale of the losses remain part of the problem.

IN-CLASS CHECK

Prompt: A rule sees f_t first and then chooses its minimizer in $\mathcal{K}$. Is it an instance of the declared protocol? If an adaptive adversary selects f_t after seeing x_t, which sequence must the comparator use?

Hint: Read the order in the definition and resist replacing the realized loss sequence for the comparator.

Answer:

Choosing after seeing f_t violates the declared protocol. An adaptive loss chosen after the commitment is permitted here, but the comparator must be evaluated on exactly the same realized functions $f_1, \dots, f_T$. Giving the comparator a different sequence would no longer measure external regret.

We have a performance ledger but no finite expert list to reweight. One local gradient at x_t must somehow control comparison with every feasible x.

Turn one local gradient into a feasible comparison

Begin with the convex quadratic $f(x) = \frac{1}{2}\|x - a\|_2^2$ on a convex set containing $a = (1, 0)$ and $x_t = (0, 1)$. Its gradient at the current point is $\nabla f(x_t) = x_t - a = (-1, 1)$.

Result: Convex loss linearization

If f_t is differentiable and convex, then for the current feasible point x_t and every feasible comparator x,

$$f_t(x_t) - f_t(x) \leq \nabla f_t(x_t)^T (x_t - x). \tag{3}$$

This is an upper bound on the loss difference. It does not say that the gradient points toward the hindsight comparator, which is defined only from the complete sequence.

The general objects are now visible: any differentiable convex loss, its gradient at the committed point, and any feasible comparator. Summing the inequality would control regret if the inner products on the right could be organized. A raw gradient step may leave $\mathcal{K}$. For example, in the box $\mathcal{K} = [0, 1]^2$, the outside point $y = (1.3, -0.2)$ has the closest feasible point $p = (1, 0)$. We now name and define that operation before asking it to do any explanatory or computational work.

Definition: Euclidean projection

For a nonempty closed convex set $\mathcal{K} \subseteq \mathbb{R}^d$, the Euclidean projection of $y \in \mathbb{R}^d$ is the registered closest-point map

$$\Pi_{\mathcal{K}}(y) = \arg\min_{x \in \mathcal{K}} \|x - y\|_2. \tag{4}$$

The stated assumptions give a unique closest point. Projection makes an outside proposal feasible. It does not assert a universal running time.

The board now asks two different questions: how a tangent controls loss, and how the newly defined closest-point map controls distance.

Tangent lane	Value	Relation to check
$f(x_t)-f(a)$		
$\nabla f(x_t)^T(x_t-a)$		$f(x_t)-f(a)\le$
condition		Why is the inequality licensed?

Box $\mathcal{K}=[0,1]^2$	outside point	closest feasible point	feasible comparator
	$y=(1.3,-0.2)$	$p=(1,0)$	$z=(0.6,0.4)$
squared distance	$\|y-z\|_2^2=$	$\|p-z\|_2^2=$	comparison:

Statement	Job	Condition or caveat
tangent inequality	regret comparison	
closest feasible point	feasibility	
projection running time	computation	

Table 2: The tangent and projection board after both tools are defined. Loss comparison, feasibility, and computational cost occupy separate lanes.

IN-CLASS CHECK

Prompt: Complete both numerical lanes and classify the three jobs. Which mathematical condition licenses the tangent inequality?

Hint: For the projection lane, compare squared distances; for the tangent lane, calculate the two sides separately.

Answer:

Here $f(x_t)=1$, $f(a)=0$, so the loss gap is 1. The inner product is $(-1,1)^T(-1,1)=2$, hence $1\le 2$. Differentiable convexity licenses this upper bound. In the box example, $\|y-z\|_2^2=0.7^2+(-0.6)^2=0.85$, while $\|p-z\|_2^2=0.4^2+(-0.4)^2=0.32$. The tangent statement serves regret comparison; the closest-point operation serves feasibility; and running time is a separate computational question determined by how $\mathcal{K}$ is represented.

Let $p=\Pi_{\mathcal{K}}(y)$ and take any feasible $z\in\mathcal{K}$. Since the entire segment $p+\lambda(z-p)$ for $\lambda\in[0,1]$ is feasible, closest-point optimality makes the right derivative at $\lambda=0$ nonnegative:

$$0\le 2(p-y)^T(z-p),\quad \text{so}\quad (y-p)^T(z-p)\le 0. \tag{5}$$

Expanding from y through p gives

$$\|y-z\|_2^2=\|y-p\|_2^2+\|p-z\|_2^2+2(y-p)^T(p-z)\ge\|p-z\|_2^2. \tag{6}$$

Taking square roots proves the distance statement used later.

Result: Projection cannot increase distance to a feasible comparator

If $\mathcal{K}$ is nonempty, closed, and convex, then for every $y \in \mathbb{R}^d$ and every $z \in \mathcal{K}$,

$$\|\Pi_{\mathcal{K}}(y) - z\|_2 \le \|y - z\|_2. \tag{7}$$

Feasibility of z is essential. A nearest point in a nonconvex set need not be unique and does not inherit this theorem from the definition alone.

IN-CLASS CHECK

Prompt: Put the following proof steps in order: expand a squared distance; use closest-point optimality along a feasible segment; take square roots; define $p = \Pi_{\mathcal{K}}(y)$. Then state where convexity of the set is used.

Hint: The segment from p toward the comparator must remain feasible before a directional derivative can be invoked.

Answer:

> First define $p = \Pi_{\mathcal{K}}(y)$. Second, use the feasible segment from p toward z to obtain $(y-p)^T(z-p) \le 0$. Third, expand $\|y - z\|_2^2$ through p and use that sign. Finally take square roots. Convexity is used to keep $p + \lambda(z - p)$ inside $\mathcal{K}$.

Worked example: Project onto a box, then audit the cost claim

For $\mathcal{K} = [0,1]^2$ and $y = (1.3, -0.2)$, coordinate clipping gives $\Pi_{\mathcal{K}}(y) = (1, 0)$. For $z = (0.6, 0.4)$, the calculation above verifies the distance theorem. Coordinate clipping happens to be cheap for a box. That example cannot support the general claim that projection onto every convex set is linear-time; a different representation of $\mathcal{K}$ can make the closest-point problem substantially harder.

Convexity has replaced a nonlinear loss gap by an inner product. Projection has made a proposed point feasible while preserving comparison distance. The remaining question is which proposed point turns those distances into a telescoping sum.

Make squared distance telescope

Start on the box $\mathcal{K} = [0,1]^2$ at $x_t = (0.9, 0.1)$. Suppose the revealed loss has gradient $g_t = (1, -1)$ and choose $\eta = 1$. The additive proposal is $y_{t+1} = x_t - \eta g_t = (-0.1, 1.1)$, outside the box; its projection is $x_{t+1} = (0, 1)$. This one round shows both operations, but a regret theorem must work for every round and every feasible fixed comparator.

Definition: Projected online gradient descent

Let $\eta_t > 0$. After choosing x_t and observing the differentiable convex loss f_t, set $g_t = \nabla f_t(x_t)$ and update with the registered form

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t)). \tag{8}$$

This is projected online gradient descent (OGD). The current revealed loss produces the update; projection makes the next decision feasible.

Card	Mathematical move	Relation to recover
1	convex loss gap	$f_t(x_t) - f_t(x) \le$
2	projected update	$x_{t+1} =$
3	projection distance	$\lVert x_{t+1} - x \rVert_2^2 \le$
4	squared-distance expansion	$g_t^T(x_t - x) \le$
5	sum over t	the distance differences

Parameter gate	Legal route	Conclusion
$D > 0$ and $G > 0$	$\eta =$	two terms balance
$D = 0$		
$G = 0$		
larger dimension d	inspect D and G	

No.	Tutorial 9B handoff slot	Reconstruction cue
1-4	four Chapter 9 entry items	
5-9	five Tutorial 9A core concepts	
10-13	four registered forms	

Table 3: The OGD proof ladder, parameter gate, and thirteen-slot boundary tray. Each card has one mathematical job.

IN-CLASS CHECK

Prompt: Order the five proof cards and fill the projected update. Why must the loss-gap card precede telescoping?

Hint: Regret is a sum of loss gaps, while squared distance directly controls only gradient inner products.

Answer:

The order is convex loss gap, projected update, projection distance, squared-distance expansion, then summation. The update is $x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta g_t)$ with $g_t = \nabla f_t(x_t)$. Linearization must come first because regret contains $f_t(x_t) - f_t(x)$, not squared distances. It converts each loss gap to the inner product that the distance expansion can bound.

Worked example: Execute one projected OGD round

In the box example, $x_t = (0.9, 0.1)$, $g_t = (1, -1)$, and $\eta_t = 1$. The raw move is $(-0.1, 1.1)$ and the registered projection returns $(0, 1)$. For the feasible comparator $x = (0.4, 0.6)$, the squared distances are $\|x_t - x\|_2^2 = 0.5$ and $\|x_{t+1} - x\|_2^2 = 0.32$. This numerical decrease is useful, but the proof below does not assume that distance decreases on every round; the gradient-size term pays for possible increases.

Fix any comparator $x \in \mathcal{K}$ and use a constant $\eta > 0$. Projection gives

$$\|x_{t+1} - x\|_2^2 \le \|x_t - \eta g_t - x\|_2^2. \tag{9}$$

Expand the right side:

$$\|x_t - x\|_2^2 - 2\eta g_t^T (x_t - x) + \eta^2 \|g_t\|_2^2. \tag{10}$$

Move the inner product to the left and divide by the positive number 2η:

$$g_t^T (x_t - x) \le \frac{\|x_t - x\|_2^2 - \|x_{t+1} - x\|_2^2}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2. \tag{11}$$

Convex linearization places the loss gap below the same right side. Summing from $t = 1$ to T makes the distance differences cancel:

$$\sum_{t=1}^{T} (f_t(x_t) - f_t(x)) \le \frac{\|x_1 - x\|_2^2 - \|x_{T+1} - x\|_2^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{T} \|g_t\|_2^2. \tag{12}$$

The final squared distance is nonnegative. If the diameter of $\mathcal{K}$ is at most D and every feasible gradient satisfies $\|\nabla f_t(z)\|_2 \le G$ for every t and $z \in \mathcal{K}$, then in particular $\|g_t\|_2 \le G$ on each played round, so

$$\sum_{t=1}^{T} (f_t(x_t) - f_t(x)) \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}. \tag{13}$$

This holds for every feasible x. In the stated OGD setting, finite diameter makes the nonempty closed set $\mathcal{K}$ bounded and therefore compact in $\mathbb{R}^d$. Differentiability makes every

f_t continuous, so their finite sum attains its infimum on $\mathcal{K}$. We may therefore choose $x^* \in \arg\min_{x\in\mathcal{K}} \sum_{t=1}^{T} f_t(x)$ and apply the bound to this best fixed comparator.

IN-CLASS CHECK

Prompt: Starting from the projected update, derive the one-step inequality for $g_t^T(x_t - x)$. Circle the sign that would break the proof if projection increased comparator distance.

Hint: Expand $\|x_t - \eta g_t - x\|_2^2$ before rearranging, and divide only after stating $\eta > 0$.

Answer:

Projection yields $\|x_{t+1} - x\|_2^2 \le \|x_t - x\|_2^2 - 2\eta g_t^T(x_t - x) + \eta^2\|g_t\|_2^2$. Rearranging gives $g_t^T(x_t - x) \le \frac{\|x_t - x\|_2^2 - \|x_{t+1} - x\|_2^2}{2\eta} + \left(\frac{\eta}{2}\right)\|g_t\|_2^2$. Reversing the projection inequality would reverse the useful control and destroy the upper-bound chain.

IN-CLASS CHECK

Prompt: Write the first three distance numerators before and after summation. Which terms cancel, and which endpoint is safely discarded?

Hint: Use $a_t = \|x_t - x\|_2^2$ and expand $(a_1 - a_2) + (a_2 - a_3) + (a_3 - a_4)$.

Answer:

The sum is $(a_1 - a_2) + (a_2 - a_3) + (a_3 - a_4) = a_1 - a_4$. All interior terms cancel. Since $a_4 \ge 0$, dropping $-a_4$ preserves an upper bound. In T rounds the same calculation leaves $\|x_1 - x\|_2^2 - \|x_{T+1} - x\|_2^2$.

Result: Constant-step OGD regret bound

Let $T \ge 1$, let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex with diameter at most D, and let $f_1, \ldots, f_T$ be differentiable convex losses. Suppose $\|\nabla f_t(z)\|_2 \le G$ for every round t and every feasible $z \in \mathcal{K}$. For every constant $\eta > 0$, projected OGD satisfies

$$R_T \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}. \tag{14}$$

If $D > 0$ and $G > 0$, the legal choice $\eta = \frac{D}{G\sqrt{T}}$ balances the two terms and gives

$$R_T \le DG\sqrt{T}. \tag{15}$$

The cases $D = 0$ and $G = 0$ are handled separately below; the displayed quotient is not used in either case.

IN-CLASS CHECK

Prompt: For $D = 6$, $G = 2$, and $T = 100$, compute the balanced step and both terms in the regret bound.

Hint: Use $\eta = \frac{D}{G\sqrt{T}}$, then evaluate the diameter and gradient terms separately.

Answer:

> The step is $\eta = \frac{6}{2\cdot 10} = 0.3$. The first term is $\frac{36}{2\cdot 0.3} = 60$, and the second is $\frac{0.3\cdot 4\cdot 100}{2} = 60$. Their sum is 120, equal to $DG\sqrt{T} = 6 \cdot 2 \cdot 10 = 120$.

The zero cases do not require a limiting argument.

Case	Direct reason	Conclusion
$D = 0$	Every two points in $\mathcal{K}$ coincide, so the feasible set is a singleton.	Learner and fixed comparator choose the same point; $R_T = 0$.
$G = 0$	Every feasible gradient is zero, so each differentiable loss is constant on the convex set.	Learner and every fixed comparator have equal loss; $R_T = 0$.
$D, G > 0$	The balanced quotient is defined and positive.	$\eta = \frac{D}{G\sqrt{T}}$ gives $R_T \le DG\sqrt{T}$.

IN-CLASS CHECK

Prompt: Why is substituting $\eta = \frac{D}{G\sqrt{T}}$ invalid when $D = 0$ or $G = 0$ even though the true regret is zero? Give the direct proof for each branch.

Hint: The theorem requires $\eta > 0$; use geometry for $D = 0$ and linearization for $G = 0$.

Answer:

> If $D = 0$, the formula proposes $\eta = 0$ when $G > 0$, outside the theorem's positive-step domain; the set is a singleton, so learner and comparator coincide. If $G = 0$, the formula divides by zero. The uniform gradient bound makes every f_t constant on the convex set, so the learner and every fixed comparator incur identical cumulative loss and regret is zero.

The notation $DG\sqrt{T}$ contains no explicit d, but this does not make the guarantee dimension-free. For example, a box whose side length is fixed can have Euclidean diameter proportional to $\sqrt{d}$; a gradient norm bound can also grow with dimension. Likewise, the proof counts projection as a legal mathematical operation without claiming that every representation of $\mathcal{K}$ supports a cheap implementation.

IN-CLASS CHECK

Prompt: A family of sets has $D = \sqrt{d}$ and the losses have $G = 3$. What dimension dependence is present in the tuned OGD bound? Does the proof determine projection running time?

Hint: Substitute the geometric parameters into $DG\sqrt{T}$, then separate regret from computation.

Answer:

> The bound becomes $3\sqrt{dT}$, so dimension is present through D. The regret proof does not determine projection running time. That cost depends on the structure and representation of the feasible set and must be analyzed separately.

Standard language to carry forward

Term or notation	Meaning here	Representative form or condition
online convex optimization	Choose a feasible point before a bounded convex loss is revealed; compare with the infimum over fixed feasible points.	$R_T = \sum_t f_t(x_t) - \inf_{x\in\mathcal{K}} \sum_t f_t(x)$
convex loss linearization	Upper-bound a loss gap by a gradient inner product at the played point.	$f_t(x_t) - f_t(x) \le \nabla f_t(x_t)^T (x_t - x)$
Euclidean projection	Return the closest feasible point without increasing distance to a feasible comparator.	$\Pi_{\mathcal{K}}(y) = \arg\min_{x\in\mathcal{K}} \lVert x - y\rVert_2$
projected online gradient descent	Move against the revealed gradient and project the next decision back to the set.	$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t))$
constant-step OGD regret bound	A diameter term and a gradient term produced by telescoping squared distance.	$R_T \le \frac{D^2}{2\eta} + \eta G^2 \frac{T}{2}$

Reconstruct the exact Tutorial 9B entry

Tutorial 9B may rely on exactly thirteen items, in this order. The first four are the Chapter 9 entry just reconstructed. The next five are the core ideas established here. The final four are the registered forms that make those ideas usable without renaming symbols.

No.	Available item	Learner-visible reconstruction
1	external regret	Best one fixed point on the same realized sequence.
2	linear MWU regret theorem	Finite-simplex comparator theorem with its legal positive-rate domain.
3	convex function	Chord inequality and its differentiable first-order consequence.
4	gradient-method template	A local additive descent proposal before online timing and feasibility are added.
5	online convex optimization	Choose x_t before seeing bounded convex f_t and compare with the fixed-point infimum

No.	Available item	Learner-visible reconstruction
		over $x \in \mathcal{K}$; use a minimum only when attained.
6	convex loss linearization	$f_t(x_t) - f_t(x) \le \nabla f_t(x_t)^T(x_t - x)$.
7	Euclidean projection	Closest point in nonempty closed convex $\mathcal{K}$; distance to feasible comparators cannot increase.
8	online gradient descent	Reveal the current loss, move against its gradient, and project the next decision.
9	OGD regret theorem	$R_T \le \frac{D^2}{2\eta} + \eta G^2 \frac{T}{2}$ and the separately gated tuning.
10	$\mathcal{K} \subseteq \mathbb{R}^d$	Registered OCO decision-set form.
11	$f_t : \mathcal{K} \to \mathbb{R}$	Registered OCO loss form.
12	$\Pi_{\mathcal{K}}(y) = \arg\min_{x \in \mathcal{K}} \lVert x - y \rVert_2$	Registered Euclidean-projection form.
13	$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t))$	Registered OGD update.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct all thirteen items in order. Then explain why the projection-cost caveat and the $D = 0, G = 0$ routes are not extra interface slots.

Hint: Use the grouping 4 prior items, 5 core concepts, and 4 registered forms.

Answer:

> The order is external regret; linear MWU regret theorem; convex function; gradient-method template; online convex optimization; convex loss linearization; Euclidean projection; online gradient descent; OGD regret theorem; the registered decision set; the registered loss; the registered projection; and the registered OGD update. Projection cost and the zero branches are conditions and boundary reasoning inside the established concepts, not additional approved items.

The required core ends here. Tutorial 9B can now ask whether minimizing past losses directly is stable and how regularization changes that answer. No leader-following rule, regularized update, logarithmic-rate condition, general lower bound, or online-to-batch claim has been used in this Tutorial.

Source note

The online convex-optimization protocol, bounded convex losses, choose-before- reveal order, and fixed-point comparator follow Hazan, **Introduction to Online Convex Optimization**, second edition, Section 1.1 (printed pp. 2-3; PDF pp. 24-25). This Tutorial writes the broad benchmark as an infimum and uses a minimum only after attainment is established. The portfolio is a course-authored linear-loss instance whose minimum is attained. The historical Week 9 sheet supplies scope and motivation only; no exact continuous-grid count, universal operation cost, or dimension-free slogan is retained.

Euclidean projection and its feasible-comparator distance property follow Hazan Section 2.1.1 and Theorem 2.1 (printed pp. 19-20; PDF pp. 41-42). Convex linearization, projected OGD, and the telescoping distance proof follow Hazan Algorithm 8, equation (3.1), and Theorem 3.1 (printed pp. 42-45; PDF pp. 64-67). The constant-step specialization is derived directly from the same one-step inequality. Its conditions $T \geq 1$, $\eta > 0$, diameter D, and gradient bound G are explicit; the positive-D, G tuning and the two zero branches are kept separate.

The optional mirror-descent connection is deliberately absent. Under the approved paired plan it may appear only as one answered, zero-minute item after Tutorial 9B's complete core and course handoff, and it exports no required knowledge.

DSAA3073: Theories in Data Science

Tutorial 9B: Stable Leaders, Licensed Fast Rates, and the Boundary to Learning

150-minute core tutorial - 15-minute break - 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

If following the empirical leader can oscillate, which geometry stabilizes the decisions, when is faster regret actually licensed, and what does online regret say about population risk?

Tutorial 9A replaced a finite expert list by a continuous convex decision set and proved the general projected-OGD square-root guarantee. That proof moved from the current point. A different instinct is to minimize all past losses and play their leader. The instinct is sensible, but the leader can jump from one boundary point to the other after a tiny change in the past total.

We will diagnose that jump on one signed loss ledger, add a strongly convex regularizer to the past **linearized** losses, and prove the resulting FTRL tradeoff. The proof will tell us exactly what regularizer geometry buys. It will not license a logarithmic rate until additional curvature enters in the losses or a matching fast-rate algorithm is supplied. A stationary random linear-loss construction then shows why the general square-root scale is unavoidable. Finally, we will keep sequential regret and distributional population risk in separate columns and state every bridge assumption needed to relate them.

Two Chapter 1 tools return at their use points. In the twice-differentiable Euclidean case, Hessian lower and upper bounds respectively certify strong convexity and gradient-Lipschitz L-smoothness; Hazan assumes both. Quantified Big-O states what grows and stays fixed. These callbacks remain outside Tutorial 9A's exact handoff.

The three core blocks take **55, 65, and 30 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge. One mirror-descent connection is sealed in a fully answered zero-minute appendix after the complete core and exact course handoff.

Clock	Mathematical work	Protected time
0-55	Break an empirical leader and build linearized FTRL	55 min core
55-70	Scheduled break	15 min
70-135	Prove the FTRL tradeoff and audit fast-rate assumptions	65 min core
135-165	Match the lower bound and separate online regret from batch risk	30 min core
165-225	Questions, reconstruction, alternative calculations	60 min buffer

When the past leader will not stay put

Tutorial 9B receives exactly thirteen items from the verified Tutorial 9A boundary. Reconstruct them before introducing a leader-following rule.

No.	Available item	Reconstruction cue
1	external regret	Compare with one fixed point on the same realized sequence.
2	linear MWU regret theorem	Recall a finite-simplex comparator theorem with a legal positive rate.
3	convex function	Use the chord inequality and its first-order consequence under its conditions.
4	gradient-method template	Recognize a local additive descent proposal.
5	online convex optimization	Choose x_t before bounded convex f_t is revealed and use a fixed-point infimum benchmark.
6	convex loss linearization	$f_t(x_t) - f_t(x) \le \nabla f_t(x_t)^T (x_t - x)$.
7	Euclidean projection	Return the closest point in nonempty closed convex $\mathcal{K}$.
8	online gradient descent	Move after the current loss is revealed, then project the next decision.
9	OGD regret theorem	$R_T \le \frac{D^2}{2\eta} + \eta G^2 \frac{T}{2}$ with separate tuning conditions.
10	$\mathcal{K} \subseteq \mathbb{R}^d$	Registered OCO decision-set form.
11	$f_t : \mathcal{K} \to \mathbb{R}$	Registered OCO loss form.
12	$\Pi_{\mathcal{K}}(y) = \arg\min_{x\in\mathcal{K}} \|x - y\|_2$	Registered Euclidean-projection form.
13	$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t))$	Registered OGD update.

The thirteen rows are exactly the Tutorial-9A interface. Chapter 1′s Hessian and Big-O definitions remain available as course-level prerequisites, but they do not enlarge or reorder this handoff.

IN-CLASS CHECK

Prompt: Write the thirteen items in order without looking back. Which item lets us replace an original convex loss gap by a linear expression? Which item guarantees only a square-root rate in the general bounded setting?

Hint: Use the grouping 4 prior items, 5 Tutorial 9A concepts, and 4 registered forms.

Answer:

> The exact order is external regret; linear MWU regret theorem; convex function; gradient-method template; online convex optimization; convex loss linearization; Euclidean projection; online gradient descent; OGD regret theorem; the registered decision set; the registered loss; registered Euclidean projection; and the registered OGD update. Convex loss linearization supplies the linear upper bound. The general OGD theorem supplies the square-root guarantee after its legal tuning.

Now work on $\mathcal{K} = [-1, 1]$. The first revealed loss is $f_1(x) = \frac{x}{2}$. For $t \ge 2$, the losses alternate:

$$f_t(x) = \begin{cases} -x & \text{if } t \text{ is even} \\ x & \text{if } t \text{ is odd} \end{cases}. \tag{16}$$

Before naming the rule, let Rule A choose a minimizer of the cumulative losses revealed strictly before the current round. If the past total is tied at round 1, choose $x_1 = 1$. Complete the ledger.

t	past total	Rule A x_t	revealed f_t	$f_t(x_t)$
1	0	1 by tie rule	$\frac{x}{2}$	
2	$\frac{x}{2}$		$-x$	
3	$-\frac{x}{2}$		x	
4	$\frac{x}{2}$		$-x$	
5	$-\frac{x}{2}$		x	

Objective card	Past tilt	Added bowl	Predicted movement
Rule A	$\sum_{s<t} g_s x$	none	endpoint to endpoint
Rule B	$\eta \sum_{s<t} g_s x$	$\frac{x^2}{2}$	

Claim	Accept?	Reason or repair
a stable update automatically has logarithmic regret		
a regularizer and population risk use the same symbol R		
the current gradient may enter x_t before x_t is played		

Table 4: One signed ledger exposes leader oscillation. The quadratic bowl changes the movement scale, but the rate still requires a theorem.

IN-CLASS CHECK

Prompt: Complete Rule A through round 5. For general T, compute its learner loss, the best fixed loss, and regret.

Hint: The cumulative past coefficient alternates between $\frac{1}{2}$ and $-\frac{1}{2}$; minimize a linear function on $[-1, 1]$.

Answer:

> The decisions are $x_1 = 1, x_2 = -1, x_3 = 1, x_4 = -1, x_5 = 1$. The first incurred loss is $\frac{1}{2}$ and every later incurred loss is 1, so the learner loses $T - \frac{1}{2}$. The complete cumulative loss is either $\frac{x}{2}$ or $-\frac{x}{2}$ and its minimum on $[-1, 1]$ is $-\frac{1}{2}$. Hence the regret is exactly T. The sign calculation, not the name of the rule, establishes the linear-regret example.

Definition: Follow the Leader

At round t, **Follow the Leader (FTL)** chooses

$$x_t \in \arg\min_{x \in \mathcal{K}} \sum_{s=1}^{t-1} f_s(x). \tag{17}$$

A tie-breaking rule is part of a fully specified implementation. The loss f_t is absent because the learner must commit before that loss is revealed. On the signed ledger above, each new loss moves the past minimizer to the opposite endpoint, so FTL is unstable and suffers linear regret.

IN-CLASS CHECK

Prompt: A proposed update minimizes $\sum_{s=1}^{t} f_s(x)$ and calls the result x_t. What protocol error has occurred? Does changing only the tie rule repair the signed example after round 1?

Hint: Compare the summation limit with the choose-before-reveal order.

Answer:

> The update uses f_t before playing x_t, so it violates the online information order. A different initial tie rule changes at most the first loss. From round 2 onward the past total has a nonzero coefficient and its minimizer alternates between endpoints, so tie-breaking alone does not repair the instability.

The failed rule has no resistance to a small change in cumulative loss. To see what resistance means, compare $A(x) = |x|$ with $B(x) = \frac{x^2}{2}$ near the origin. The first graph has a corner and linear sides; the second has a quadratic bowl. Nonsmoothness and curvature are different questions.

Definition: Strong convexity

Relative to a declared norm $\|\cdot\|$, a differentiable function $\mathcal{R} : \mathcal{K} \to \mathbb{R}$ is α-strongly convex when $\alpha > 0$ and for every $x, y \in \mathcal{K}$,

$$\mathcal{R}(y) \geq \mathcal{R}(x) + \nabla \mathcal{R}(x)^T (y - x) + \frac{\alpha}{2}\|y - x\|^2. \tag{18}$$

For a nonsmooth convex function, the same inequality is stated with any valid subgradient at x. Strong convexity supplies a quadratic lower gap. It is not the definition of smoothness, differentiability, or a unique gradient. On $[-1, 1]$, $\mathcal{R}(x) = \frac{x^2}{2}$ is 1-strongly convex in the Euclidean norm.

For a twice-differentiable function on a convex Euclidean domain, Hazan's assumptions read

$$\alpha I \preccurlyeq \nabla^2 \mathcal{R}(x) \preccurlyeq \beta I \quad \text{for every} \quad x. \tag{19}$$

Under the usual interior regularity conditions, the lower bound is equivalent to the first-order inequality and certifies strong convexity; the upper bound certifies Chapter 1 β-smoothness. Neither replaces the general subgradient definition, which also covers nonsmooth strongly convex functions.

IN-CLASS CHECK

Prompt: Classify the statements: (i) $|x|$ is nonsmooth, therefore strongly convex; (ii) $|x| + \alpha\frac{x^2}{2}$ is nonsmooth and α-strongly convex; (iii) the constant function is smooth and strongly convex.

Hint: Test the quadratic lower gap separately from differentiability.

Answer:

> Statement (i) is false: $|x|$ has linear pieces and no positive quadratic lower gap on an interval. Statement (ii) is true: adding the quadratic supplies the required strong convexity even though the corner remains. Statement (iii) is false: a constant has a Lipschitz gradient but no positive curvature parameter. Thus nonsmoothness neither implies nor forbids strong convexity.

Tutorial 9A already gave the legal linear upper model $g_s^T x$, where $g_s = \nabla f_s(x_s)$. We can therefore stabilize the past linearized losses while preserving the online information boundary.

Definition: Follow the Regularized Leader

Given $\eta > 0$ and a strongly convex regularizer $\mathcal{R}$, the registered **Follow the Regularized Leader (FTRL)** update is

$$x_t = \arg\min_{x\in\mathcal{K}} \left[\eta \sum_{s=1}^{t-1} g_s^T x + \mathcal{R}(x)\right], \quad g_s = \nabla f_s(x_s). \tag{20}$$

Only past linearized gradients occur in x_t. The calligraphic symbol $\mathcal{R}$ denotes the regularizer; plain $R(h)$ remains available for population risk. In Hazan this linearized form is called Regularized Follow the Leader (RFTL).

For the signed ledger, let $\mathcal{R}(x) = \frac{x^2}{2}$ and $0 < \eta \le 2$. If $S_{t-1} = \sum_{s<t} g_s$, then $x_t = \mathrm{clip}(-\eta S_{t-1}, -1, 1)$. Hence $x_1 = 0$ and the later decisions alternate between $-\frac{\eta}{2}$ and $\frac{\eta}{2}$. FTL jumped a distance of 2; this FTRL example jumps only η. With a fixed η, however, the repeated incurred losses remain of order ηT. Stability is visible. A logarithmic theorem is not.

IN-CLASS CHECK

Prompt: For $\eta = \frac{1}{2}$, compute x_1, x_2, x_3, x_4 on the signed ledger. Which exact change creates stability, and which rate claim is still unjustified?

Hint: Use $S_0 = 0, S_1 = \frac{1}{2}, S_2 = -\frac{1}{2}, S_3 = \frac{1}{2}$.

Answer:

> The decisions are $x_1 = 0, x_2 = -\frac{1}{4}, x_3 = \frac{1}{4}, x_4 = -\frac{1}{4}$. The strongly convex quadratic makes the objective resist endpoint jumps and limits each move here to $\frac{1}{2}$. This calculation does not prove logarithmic regret; with a fixed step, even this example accumulates a loss contribution of order ηT. A regret theorem and a legal choice of η are still needed.

The unresolved question is now precise: how does a bound on consecutive FTRL movement enter regret, and what rate follows when the two resulting terms are balanced?

Geometry controls stability; assumptions control the rate

All asymptotic rate symbols in this block reuse Chapter 1′s quantified Big-O definition with horizon $T \to \infty$. The decision-set diameter, gradient bounds, curvature parameters, dimension, and any other declared problem parameters are fixed unless a theorem says otherwise. Thus $R_T = O\left(\sqrt{T}\right)$, for example, means that constants $C > 0$ and T_0 exist such that $R_T \le C\sqrt{T}$ for every $T \ge T_0$; C may depend on the declared fixed problem parameters but not on T. The same convention governs $O(\log T)$. These rate classes suppress constants—they do not assert equality or make two theorems interchangeable.

For a fixed comparator $u \in \mathcal{K}$, write $\Delta_u = \mathcal{R}(u) - \mathcal{R}(x_1) \ge 0$, where x_1 minimizes $\mathcal{R}$ on $\mathcal{K}$. The proof first separates regret into a movement bill and a comparator-geometry bill:

$$R_T(u) \le \sum_{t=1}^{T} g_t^T (x_t - x_{t+1}) + \frac{\Delta_u}{\eta}. \tag{21}$$

To measure the movement in the regularizer's geometry, define its first-order remainder $B_{\mathcal{R}}(x, y) = \mathcal{R}(x) - \mathcal{R}(y) - \nabla\mathcal{R}(y)^T (x - y)$. Under Hazan's twice-differentiable, α-strongly-convex, β-smooth setup—smooth in Chapter 1′s Lipschitz-gradient sense—the remainder between x_t and x_{t+1} is $B_{\mathcal{R}}(x_t, x_{t+1}) = \frac{\|x_t - x_{t+1}\|_t^2}{2}$ for a local norm induced by the regularizer curvature. Its dual norm is $\|\cdot\|_t^*$. These are proof coordinates, not a renamed global Euclidean norm.

Step	Proof card	Mathematical job
1	convex linearization	Replace $f_t(x_t) - f_t(u)$ by $g_t^T(x_t - u)$.
2	leader comparison	Convert the cumulative linearized comparator gap into successive movement plus $\frac{\Delta_u}{\eta}$.
3	regularizer curvature	Bound $B_{\mathcal{R}}(x_t, x_{t+1})$ by $\eta g_t^T(x_t - x_{t+1})$.
4	local dual Cauchy-Schwarz	Turn the movement inner product into $2\eta\left(\|g_t\|_t^*\right)^2$.

Assumption row	Geometry or curvature	Matching schedule or algorithm	Licensed rate
general convex losses	fixed strong $\mathcal{R}$	fixed-horizon FTRL balance	$O\left(\sqrt{T}\right)$
strongly convex losses	negative quadratic in every f_t	$\eta_t = \frac{1}{\alpha t}$ OGD	$O(\log T)$
exp-concave losses	$\exp(-\alpha f_t)$ concave	matching EWOO or second-order method	$O(\log T)$

Loose theorem card	Place?	Required repair
fixed strongly convex regularizer $\to$ $O(\log T)$ for arbitrary convex losses		
use $\eta_t = \frac{1}{\alpha t}$ without strong loss curvature		
exp-concavity $\to$ every first-order update is logarithmic		

Table 5: The proof ladder and rate-evidence matrix keep regularizer geometry, loss curvature, schedule, and algorithm in separate slots.

Result: FTRL regret tradeoff

Let $\mathcal{K}$ be bounded, closed, and convex. Let the losses be convex and let the regularizer satisfy Hazan's twice-differentiable, α-strongly-convex, β-smooth setup. For the registered linearized FTRL update and every $u \in \mathcal{K}$,

$$R_T(u) \le 2\eta \sum_{t=1}^{T} \left(\|g_t\|_t^*\right)^2 + \frac{\Delta_u}{\eta}. \tag{22}$$

If $\|g_t\|_t^* \le G_{\mathcal{R}}$ and a known uniform bound $\Delta_{\mathcal{R}}$ satisfies $\mathcal{R}(u) - \mathcal{R}(x_1) \le \Delta_{\mathcal{R}}$ for every $u \in \mathcal{K}$, then

$$R_T(u) \le 2\eta G_{\mathcal{R}}^2 T + \frac{\Delta_{\mathcal{R}}}{\eta}. \tag{23}$$

When $\Delta_{\mathcal{R}} > 0$ and $G_{\mathcal{R}} > 0$, the fixed-horizon choice $\eta = \sqrt{\frac{\Delta_{\mathcal{R}}}{2G_{\mathcal{R}}^2 T}}$ balances the two terms and gives order $\sqrt{T}$. The theorem does not turn arbitrary convex losses into a logarithmic-regret class.

Here is the decisive movement inequality. Optimality of consecutive regularized leaders gives $B_{\mathcal{R}}(x_t, x_{t+1}) \le \eta g_t^T(x_t - x_{t+1})$. Local dual Cauchy-Schwarz then yields

$$g_t^T(x_t - x_{t+1}) \le \|g_t\|_t^* \|x_t - x_{t+1}\|_t \le \|g_t\|_t^* \sqrt{2\eta g_t^T(x_t - x_{t+1})}. \tag{24}$$

Rearranging gives $g_t^T(x_t - x_{t+1}) \le 2\eta\left(\|g_t\|_t^*\right)^2$. Substitution into the movement bill proves the displayed tradeoff. The regularizer controls the successive-decision term; the comparator pays through $\frac{\Delta_u}{\eta}$.

IN-CLASS CHECK

Prompt: Order the four proof cards and justify the last rearrangement. Which term records comparator geometry, and which term records instability?

Hint: Call the nonnegative inner product a and solve $a \le b\sqrt{a}$.

Answer:

> The order is convex linearization, leader comparison, regularizer curvature, and local dual Cauchy-Schwarz. With $a = g_t^T(x_t - x_{t+1})$ and $b = \sqrt{2\eta}\|g_t\|_t^*$, the inequality $a \le b\sqrt{a}$ gives $a \le b^2 = 2\eta\left(\|g_t\|_t^*\right)^2$ (the $a = 0$ case is immediate). The term $\frac{\Delta_u}{\eta}$ records comparator geometry; the accumulated local-dual term records the movement or instability bill.

IN-CLASS CHECK

Prompt: Suppose the known uniform comparator-gap bound is $\Delta_{\mathcal{R}} = \frac{1}{2}$, $G_{\mathcal{R}} = 1$, and $T = 100$. Balance the two theorem terms. What bound results, and why is this not logarithmic?

Hint: Minimize $2\eta T + \frac{\frac{1}{2}}{\eta}$ over positive η.

Answer:

> The balanced step is $\eta = \sqrt{\frac{\frac{1}{2}}{2\cdot 100}} = \frac{1}{20}$. Each term equals 10, so the bound is 20. In general the balanced expression is proportional to $\sqrt{T}$. The strong regularizer supplied the stability inequality, but no additional curvature in the arbitrary convex losses has produced the harmonic cancellation required for $\log T$.

Result: Conditions that license logarithmic regret

A logarithmic rate requires an additional proved structure-and-algorithm match. Two standard routes are:

- If every loss f_t is α-strongly convex, gradients have norm at most G, and projected OGD uses $\eta_t = \frac{1}{\alpha t}$, then
$$R_T \le \frac{G^2}{2\alpha}(1 + \log T). \tag{25}$$
- A loss is α-exp-concave when $x \mapsto \exp(-\alpha f(x))$ is concave on the decision set. Exp-concave losses admit logarithmic regret with a matching method such as exponentially weighted online optimization or an appropriate second-order algorithm.

A fixed strongly convex regularizer alone does not imply $O(\log T)$ regret for arbitrary convex losses, and writing a decreasing step without the matching proof does not create the missing curvature.

The first route can be located in one line of the OGD proof. Strong convexity of each **loss** contributes

$$2(f_t(x_t) - f_t(u)) \le 2g_t^T(x_t - u) - \alpha\|x_t - u\|_2^2. \tag{26}$$

The negative quadratic term is absent for a merely convex loss. Combining it with the projected-distance inequality and choosing $\eta_t = \frac{1}{\alpha t}$ cancels the successive distance coefficients; only $\sum_{t=1}^{T} \frac{1}{t} \le 1 + \log T$ remains. The harmonic sum is therefore a consequence of loss curvature plus the matching schedule, not a slogan attached to regularization.

IN-CLASS CHECK

Prompt: Place each card in the only licensed row: (a) fixed strong regularizer and arbitrary convex losses; (b) strongly convex losses with $\eta_t = \frac{1}{\alpha t}$ OGD; (c) exp-concave losses with a matching second-order method. Then repair the three loose theorem cards.

Hint: A fast-rate statement needs both its curvature source and its matching proof mechanism.

Answer:

Card (a) belongs in the general FTRL square-root row. Card (b) belongs in the strongly convex loss row and uses the negative quadratic term plus the harmonic schedule. Card (c) belongs in the exp-concave row and needs its matching algorithm. The repaired statements are: a fixed strong regularizer gives stability and a general square-root tradeoff; the decreasing OGD step is licensed by strong loss curvature; and exp-concavity does not make every first-order update logarithmic.

IN-CLASS CHECK

Prompt: Where exactly does $\sum_t \frac{1}{t}$ enter the strongly convex OGD proof? Why can the same line not be copied into the general FTRL row?

Hint: Identify the negative quadratic term before invoking the harmonic bound.

Answer:

Strong convexity of each f_t subtracts $\alpha\frac{\|x_t-u\|^2}{2}$ from the first-order upper bound. With $\eta_t = \frac{1}{\alpha t}$, that term cancels the neighboring distance coefficients, leaving a gradient contribution proportional to $\sum_t \eta_t = \left(\frac{1}{\alpha}\right)\sum_t \frac{1}{t}$. The general FTRL row assumes only convex losses, so this negative loss-curvature term is unavailable; copying the harmonic line would leave an uncancelled distance sum.

We have now separated the rates that are achievable under added structure. The remaining question runs in the opposite direction: can any method beat the square-root scale when the bounded convex-loss class is left unrestricted?

What cannot be improved, and what can be transferred

Let $\mathcal{K} = \{x \in \mathbb{R}^n : \|x\|_\infty \le 1\}$. On every round draw v_t independently and uniformly from the 2^n sign vectors $\{-1, 1\}^n$, and reveal the linear loss $f_t(x) = v_t^T x$. This is one fixed stationary distribution over bounded linear losses.

Step	Online learner column	Fixed comparator column	Scale
1	x_t depends only on past signs	x may use all $v_1, \dots, v_T$	information split
2	$\mathbb{E}[v_t^T x_t \mid \text{past}] = 0$	$\min_{x\in\mathcal{K}} \sum_t v_t^T x = -\sum_i \left\lvert \sum_t v_t(i) \right\rvert$	mean losses
3	$\mathbb{E} \sum_t f_t(x_t) = 0$	$\mathbb{E} \min_x \sum_t f_t(x) = -\Omega\left(n\sqrt{T}\right)$	expected regret
4	$D \le 2\sqrt{n}$	$G \le \sqrt{n}$	$\Omega\left(DG\sqrt{T}\right)$

Online object	Bridge assumptions	Batch object
sequence losses $f_1, \dots, f_T$	iid labeled examples	one distribution $\mathcal{D}$
fixed-comparator regret R_T	bounded convex loss and convex class	population risk $R(h)$
iterates $h_1, \dots, h_T$	average $\bar{h} = T^{-1} \sum_t h_t$	one returned hypothesis
$\frac{\text{Regret}_T(h)}{T}$	concentration for adaptive iterates and comparator	high-probability sampling deviation

Claim to audit	Missing item	Repaired claim
sublinear regret proves population generalization		
$\frac{\text{Regret}_T(h)}{T}$ and $R(h)$ are the same object		
averaging works for every hypothesis class and loss		

Table 6: A stationary lower-bound construction sits beside the online-to-batch bridge. The two arguments use probability for different jobs.

Result: General square-root regret lower bound

For every online convex-optimization algorithm, there is a bounded convex decision problem with bounded linear gradients for which worst-case regret is

$$\Omega\Big(DG\sqrt{T}\Big). \tag{27}$$

Independent losses drawn from one fixed stationary distribution already suffice for the construction. The result matches the general OGD and FTRL square-root scales up to constants and problem parameters. It does not say that faster rates are impossible after strong loss curvature, exp-concavity, or another separately stated restriction is added.

The learner column has conditional expectation zero because v_t is a fresh mean-zero sign vector after x_t is fixed. The hindsight comparator chooses each coordinate opposite the completed random walk, giving $-\sum_i\Big|\sum_t v_t(i)\Big|$. A length-$T$ Rademacher walk has expected absolute displacement of order $\sqrt{T}$, so the comparator gains order $n\sqrt{T}$. Since the Euclidean diameter and gradient bound are $D \le 2\sqrt{n}$ and $G \le \sqrt{n}$, this is the required $DG\sqrt{T}$ scale.

IN-CLASS CHECK

Prompt: Complete the expectation sketch. Why is the loss distribution stationary, and why does the result not contradict the logarithmic rows above?

Hint: Condition on the past for the learner; optimize each comparator coordinate after all signs are known.

Answer:

The same uniform distribution over sign vectors is used independently on every round, so it is fixed and stationary. Conditional on the past, x_t is fixed and $\mathbb{E}[v_t] = 0$, hence the learner's expected current loss is zero. The best fixed comparator chooses $x(i) = -\mathrm{sign}\Big(\sum_t v_t(i)\Big)$ and has expected loss $-\Omega\Big(n\sqrt{T}\Big)$. The logarithmic rows impose additional curvature and algorithm conditions; the lower bound ranges over the unrestricted bounded OCO class, so there is no contradiction.

Result: A valid online-to-batch bridge

Sequence-wise regret and distributional population risk are different objects. One valid bridge assumes iid examples $(z_t)_{t=1}^{T}$ from one distribution, a convex decision or hypothesis class, a bounded loss $\ell \in [0,1]$ that is convex in the decision, online iterates h_t chosen before z_t, the average $\overline{h} = T^{-1}\sum_t h_t$, and concentration for the adaptive sequence and the fixed comparator.

For an arbitrary fixed comparator h in the class, define its comparator-specific regret by

$$\mathrm{Regret}_T(h) = \sum_{t=1}^{T} \ell(h_t, z_t) - \sum_{t=1}^{T} \ell(h, z_t). \tag{28}$$

Under Hazan's reduction, with probability at least $1-\delta$,

$$R\big(\overline{h}\big) \le R(h) + \frac{\mathrm{Regret}_T(h)}{T} + \sqrt{\frac{8\log\left(\frac{2}{\delta}\right)}{T}}. \tag{29}$$

Thus comparator-specific sublinear regret contributes the term $\frac{\mathrm{Regret}_T(h)}{T}$ only after the sampling, convexity, averaging, boundedness, and concentration assumptions have been supplied.

The role of convexity is visible: it gives $R\big(\overline{h}\big) \le T^{-1}\sum_t R(h_t)$. The iid assumption makes the loss on the fresh example an unbiased observation of each already chosen h_t's risk. Boundedness supports the concentration step. None of these facts follows from the symbol $\mathrm{Regret}_T(h)$ alone.

IN-CLASS CHECK

Prompt: Repair the three claims in the bridge matrix. Then state the five assumptions that must appear before the high-probability population-risk conclusion.

Hint: Keep the sequence, distribution, iterate average, and deviation term in different columns.

Answer:

> Sublinear regret alone is a statement about a realized loss sequence, not population generalization. The normalized comparator-specific regret $\frac{\mathrm{Regret}_T(h)}{T}$ is one term in a reduction, whereas $R(h)$ is an expectation under a distribution. Averaging is licensed by a convex class and convex loss, not universally. The required assumptions are iid examples, bounded convex loss, a convex decision or hypothesis class, the averaged iterate, and a concentration argument. The online iterate must also be chosen before its current sample.

The course has used related potentials and comparators without declaring five different updates to be the same algorithm.

Method	Update object	Do not identify it with
Euclidean OGD	gradient step plus Euclidean projection	general FTRL without specifying matching geometry
FTRL	past linearized gradients plus $\mathcal{R}$	FTL on original nonlinear losses

Method	Update object	Do not identify it with
Hedge	optional exponential expert weights	the course-primary linear MWU update
linear MWU	$w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i))$	Hedge or AdaBoost by name alone
AdaBoost	fixed training-example distribution and weak-hypothesis coefficient	an OCO method with identical iterates

IN-CLASS CHECK

Prompt: For each method, name the object it updates and one non-equivalence that must remain visible. What conditional equivalence is deliberately absent from the core table?

Hint: Shared potential reasoning does not imply identical states, losses, or update rules.

Answer:

OGD updates a feasible continuous decision through a gradient step and Euclidean projection. FTRL minimizes past linearized gradients plus a regularizer. Hedge uses exponential expert weights; the course-primary MWU uses a linear multiplicative factor; AdaBoost updates a distribution on fixed training examples while choosing weak hypotheses and coefficients. None of these names makes the iterates identical. The conditional lazy mirror-descent/RFTL equivalence for linear costs is absent from the core and appears only in the zero-minute appendix.

Standard language to carry forward

Term or notation	Meaning here	Representative form or condition
Follow the Leader	Minimize cumulative losses revealed strictly before the current round.	$x_t \in \arg\min_x \sum_{s<t} f_s(x)$
strong convexity	A first-order lower model plus a positive quadratic gap in a declared norm.	$\mathcal{R}(y) \geq \mathcal{R}(x) + \nabla\mathcal{R}(x)^T(y-x) + \alpha\frac{\|y-x\|^2}{2}$
Follow the Regularized Leader	Minimize past linearized gradients plus strongly convex regularizer geometry.	$x_t = \arg\min_x \left(\eta\sum_{s<t} g_s^T x + \mathcal{R}(x)\right)$
FTRL regret bound	Comparator regularizer cost over η plus an η-weighted local-dual gradient sum.	$2\eta\sum_t \left(\|g_t\|_t^*\right)^2 + \frac{\Delta_u}{\eta}$
conditions for logarithmic regret	Extra loss curvature or exp-concavity together with a matching schedule or algorithm.	$\eta_t = \frac{1}{\alpha t}$ for strongly convex losses
square-root lower bound	General bounded OCO has unavoidable worst-case square-root regret.	$\Omega\left(DG\sqrt{T}\right)$
online-to-batch reduction	Add iid sampling, convex averaging, bounded loss, and concentration before concluding about risk.	$R\left(\bar{h}\right) \leq R(h) + \frac{\text{Regret}_T(h)}{T}$ + deviation

Reconstruct the exact course-completion interface

The course exits with exactly three ordered items. The online-to-batch bridge is a core synthesis skill, but it is not a fourth exit slot.

No.	Exact exit	Learner-visible reconstruction
1	OGD regret theorem	$R_T \le \frac{D^2}{2\eta} + \eta G^2 \frac{T}{2}$, with legal positive tuning and separate zero cases.
2	Follow the Regularized Leader	$x_t = \arg\min_{x \in \mathcal{K}} \left[\eta \sum_{s<t} g_s^T x + \mathcal{R}(x)\right]$ using past linearized gradients.
3	general square-root lower bound	Every OCO algorithm faces some bounded linear-loss problem with $\Omega\big(DG\sqrt{T}\big)$ regret; a stationary distribution suffices.

IN-CLASS CHECK

Prompt: Reconstruct the three exits in exact order. For each, state one validity condition or boundary. Then explain why the logarithmic-rate conditions and online-to-batch skill are not extra exit slots.

Hint: Use upper bound, stabilizing method, matching lower bound.

Answer:

> The order is the OGD regret theorem; Follow the Regularized Leader; and the general square-root lower bound. OGD needs convex differentiable losses, bounded diameter and gradients, $\eta > 0$, and separately gated tuning. FTRL uses past linearized gradients and a declared strongly convex regularizer. The lower bound concerns the unrestricted bounded OCO class and permits a fixed stationary loss distribution. Logarithmic rates require extra structure, and online-to-batch reasoning requires bridge assumptions; they qualify how results may be used rather than adding course-exit slots.

The required course core ends here. Tutorial 9A established the OCO and OGD chain; this Tutorial has established leader stability, the rate conditions, the matching lower bound, and the online-to-batch boundary. The 60-minute protected buffer may revisit any core calculation but cannot introduce a new prerequisite.

Source note

The valid signed FTL construction follows Hazan, **Introduction to Online Convex Optimization**, second edition, Chapter 5 introduction (printed p. 71; PDF p. 93). Sections 2.1 and 5.1 give strong convexity, β-smoothness, their respective Hessian bounds, and the regularizer setup; smoothness here is the Chapter 1 Lipschitz-gradient notion, not the lower-bound certificate. The registered past-linearized-gradient FTRL/RFTL update, stability lemma, and local-dual-norm tradeoff follow Algorithm 13, Theorem 5.2, and Lemmas 5.3-5.4 (printed pp. 72-77; PDF pp. 94-99).

The strongly convex loss result with $\eta_t = \frac{1}{\alpha t}$ follows Hazan Theorem 3.3 (printed pp. 47-48; PDF pp. 69-70). Exp-concavity and its matching logarithmic methods follow Chapter 4 (printed pp. 55-69; PDF pp. 77-91). The general stationary linear-loss lower bound follows Section 3.2 and Theorem 3.2 (printed pp. 45-46; PDF pp. 67-68). Arora-Hazan-Kale Section 4 and Theorem 4.1 (journal pp. 153-154; PDF pp. 33-34) supply only the corresponding discrete expert-advice comparison, not the controlling OCO theorem.

The online-to-batch construction and displayed high-probability bound follow Hazan Section 9.2, Algorithm 29, and Theorem 9.5 (printed pp. 157-162; PDF pp. 179-184). The historical Week 9 sheet supplies scope and motivation only. Its sign-reversed leader table, nonlinear FTRL substitution, fixed-regularizer logarithmic claim, lower-bound misclassification, and identical-algorithm slogans are not retained.

Optional deep dive - mirror descent and FTRL

0 class minutes; post-class or self-study. Nothing here is required later or assessed.

Question. Under which exact conditions do lazy online mirror descent and RFTL produce the same predictions, and what does that equivalence not say?

Answer. Let every cost be linear, use the same differentiable strictly convex regularizer $\mathcal{R}$ and step η, and let lazy online mirror descent accumulate gradients in the dual coordinates $\nabla\mathcal{R}(y_t) = -\eta\sum_{s<t} g_s$ before projecting with the matching regularizer divergence. Expanding that divergence gives, up to terms independent of x,

$$\mathcal{R}(x) + \eta\sum_{s<t} g_s^T x. \tag{30}$$

Its constrained minimizer is therefore exactly the registered RFTL prediction. This is Hazan's Lemma 5.5. The conclusion is conditional on linear costs and matching geometry. It does not identify agile mirror descent with RFTL for arbitrary nonlinear losses, and it does not make Euclidean OGD, FTRL, Hedge, linear MWU, or AdaBoost the same algorithm.